

# Package ‘TernTables’

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**Type** Package

**Title** Automated Statistical Analysis and Table Generation for Biomedical Research

**Version** 1.7.2

**Description** Generates publication-ready summary tables for clinical research, supporting descriptive summaries and comparisons across two or three groups. The package streamlines the analytical workflow by detecting variable types and applying appropriate statistical tests (Welch t-test, Wilcoxon rank-sum, Welch ANOVA, Kruskal-Wallis, Chi-squared, or Fisher's exact test). Results are formatted as 'tibble' objects and can be exported to 'Word' or 'Excel' using the 'officer', 'flextable', and 'writexl' packages. Optional pairwise post-hoc testing for three-group comparisons (Games-Howell and Dunn's test) is available via the 'rstatix' package. Example data are derived from the landmark adjuvant colon cancer trial described in Moertel et al. (1990) <[doi:10.1056/NEJM199002083220602](https://doi.org/10.1056/NEJM199002083220602)>.

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**URL** <https://cran.r-project.org/package=TernTables>,  
<https://github.com/jdpreston30/TernTables>,  
<https://tern-tables.com/>

**BugReports** <https://github.com/jdpreston30/TernTables/issues>

**Encoding** UTF-8

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**Imports** cli,  
dplyr (>= 1.0.0),  
epitools,  
flextable (>= 0.9.0),  
magrittr,  
multcompView,  
officer (>= 0.4.6),  
rlang,  
rstatix,  
stats,  
stringr,  
tibble,  
withr,  
writexl

**Suggests** knitr,  
rmarkdown,  
survival,  
testthat (>= 3.0.0)

**VignetteBuilder** knitr

**Depends** R (>= 4.1.0)

**LazyData** true

**Config/testthat/edition** 3

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|                    |                                                             |
|--------------------|-------------------------------------------------------------|
| classify_normality | <i>Classify variables by normality and routing decision</i> |
|--------------------|-------------------------------------------------------------|

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Description

Applies the same normality assessment logic used internally by ternG() and ternD() and returns a tidy tibble showing per-variable (and per-group) statistics, the gate that triggered the routing decision, and the final parametric / non-parametric routing outcome.

Usage

```
classify_normality(  
  data,  
  vars = NULL,  
  exclude_vars = NULL,  
  group_var = NULL,  
  consider_normality = "ROBUST"  
)
```

## Arguments

|                                 |                                                                                                                                                                                                                                                                                                |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>data</code>               | A data frame or tibble.                                                                                                                                                                                                                                                                        |
| <code>vars</code>               | Optional character vector of variable names to assess. If NULL (default), all assessable numeric variables are included (excluding <code>exclude_vars</code> and <code>group_var</code> ).                                                                                                     |
| <code>exclude_vars</code>       | Optional character vector of variable names to exclude.                                                                                                                                                                                                                                        |
| <code>group_var</code>          | Optional name of the grouping variable (as used in <code>ternG()</code> ). When provided, the normality assessment is performed across groups simultaneously — exactly as <code>ternG()</code> does. When NULL, each variable is assessed as a single vector (matching <code>ternD()</code> ). |
| <code>consider_normality</code> | Normality assessment mode — must match what was (or will be) passed to <code>ternG()</code> / <code>ternD()</code> to guarantee identical routing. "ROBUST" (default), TRUE, or FALSE.                                                                                                         |

## Details

Useful for:

- Answering reviewer questions about normality testing ("was Age normally distributed?").
- Verifying that a given variable's routing matches your expectation before running `ternG()` or `ternD()`.
- Generating a supplemental normality audit table for a manuscript.

## Value

A tibble with one row per variable  $\times$  group (or one row per variable when `group_var` = NULL), containing:

**variable** Variable name.

**group** Group level, or "[all]" when no `group_var` is supplied.

**n** Non-missing sample size in this group.

**skewness** Sample skewness (population moments).

**kurtosis** Excess kurtosis (population moments; 0 for a normal distribution).

**sw\_p** Shapiro-Wilk p-value for this group. NA when the routing decision was made at Gates 1–3 under "ROBUST", when n is outside the valid range (3–5000), or when `consider_normality` = FALSE.

**gate** Integer 1–4 indicating which gate made the routing decision under `consider_normality` = "ROBUST", or NA for TRUE / FALSE modes.

**gate\_reason** Plain-language explanation of the gate decision, naming which group(s) triggered the rule where relevant.

**is\_normal** Logical; TRUE = routed to parametric (mean  $\pm$  SD, t-test / ANOVA); FALSE = non-parametric (median [IQR], Wilcoxon / Kruskal-Wallis).

**routing** Human-readable routing summary: "Parametric (mean  $\pm$  SD)" or "Non-parametric (median [IQR])".

**Examples**

```

data(tern_colon)

# Single-group audit (ternD-style)
classify_normality(tern_colon, exclude_vars = "ID")

# Grouped audit matching a ternG call
classify_normality(tern_colon, exclude_vars = "ID", group_var = "Recurrence")

# Specific variables only
classify_normality(tern_colon,
                  vars      = c("Age", "Positive_Lymph_Nodes_n"),
                  group_var = "Recurrence")

# Using Shapiro-Wilk only (matches consider_normality = TRUE in ternG/ternD)
classify_normality(tern_colon, exclude_vars = "ID",
                  group_var      = "Recurrence",
                  consider_normality = TRUE)

```

---

```
print.ternP_result
```

*Print method for ternP\_result objects*

---

**Description**

Re-displays the preprocessing summary for a `ternP_result` object. Note that [ternP](#) already emits this summary automatically at the time it is called, so this method is most useful for reviewing the summary after the fact (e.g. typing `result` at the console later in a session).

**Usage**

```
## S3 method for class 'ternP_result'
print(x, ...)
```

**Arguments**

|                  |                                                                        |
|------------------|------------------------------------------------------------------------|
| <code>x</code>   | A <code>ternP_result</code> object returned by <a href="#">ternP</a> . |
| <code>...</code> | Currently unused; included for S3-method compatibility.                |

**Value**

Invisibly returns `x`.

---

ternB

---

Combine multiple ternD/ternG tables into a single Word document

---

## Description

Takes a list of tibbles previously created by `ternD()` or `ternG()` and writes them all into one .docx file, one table per page, preserving the exact formatting settings that were used when each table was built.

## Usage

```
ternB(
  tables,
  output_docx,
  page_break = TRUE,
  methods_doc = FALSE,
  methods_filename = "TernTables_methods.docx",
  open_doc = TRUE,
  citation = TRUE,
  font_family = getOption("TernTables.font_family", "Arial")
)
```

## Arguments

|                  |                                                                                                                                                                                                                                                                                                                                  |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| tables           | A <b>list</b> of tibbles created by <code>ternD()</code> or <code>ternG()</code> . Must be constructed with <code>list()</code> , not <code>c()</code> (e.g. <code>list(T1, T2, T3)</code> ). Each tibble must have been produced in the <i>current R session</i> ; the metadata is stored in memory, not in the tibble columns. |
| output_docx      | Output file path ending in .docx.                                                                                                                                                                                                                                                                                                |
| page_break       | Logical; if TRUE (default), inserts a page break between each consecutive table.                                                                                                                                                                                                                                                 |
| methods_doc      | Logical; if TRUE, writes a single methods section Word document that covers all tables in the list. Statistical test details are pooled across all tables. Default is FALSE.                                                                                                                                                     |
| methods_filename | Output file path for the methods document. Defaults to "TernTables_methods.docx" in the working directory.                                                                                                                                                                                                                       |
| open_doc         | Logical; if TRUE (default), automatically opens each written .docx in the system default application after saving. Set to FALSE to suppress.                                                                                                                                                                                     |
| citation         | Logical; if TRUE (default), appends a citation line at the bottom of each table footnote block and methods document: package version, authors, and links to the GitHub repository and web interface. Set to FALSE to suppress.                                                                                                   |
| font_family      | Character; font family for all Word output. Any font name accepted by the rendering system is valid. Can also be set via <code>options(TernTables.font_family = "Garamond")</code> . Default "Arial".                                                                                                                            |

## Details

ternB() works by replaying the exact word\_export() call that ternD() / ternG() would have made – using stored metadata attached as an attribute to each returned tibble – but directing all output into a single combined document instead of separate files.

Table captions (table\_caption) and footnotes (table\_footnote) specified in the original ternD() / ternG() call are reproduced automatically. You can override them by modifying the "ternB\_meta" attribute before calling ternB(), though in practice it is easier to set captions and footnotes when you first build each table.

## Value

Invisibly returns the path to the written Word file.

## Examples

```
data(tern_colon)

T1 <- ternD(tern_colon,
  exclude_vars = "ID",
  table_caption = "Table 1. Overall patient characteristics.",
  methods_doc = FALSE,
  open_doc = FALSE)

T2 <- ternG(tern_colon,
  group_var = "Recurrence",
  exclude_vars = "ID",
  table_caption = "Table 2. Characteristics by recurrence status.",
  methods_doc = FALSE,
  open_doc = FALSE)

ternB(list(T1, T2),
  output_docx = file.path(tempdir(), "combined_tables.docx"),
  open_doc = FALSE)
```

---

|       |                                                                        |
|-------|------------------------------------------------------------------------|
| ternD | <i>Generate descriptive summary table (optionally normality-aware)</i> |
|-------|------------------------------------------------------------------------|

---

## Description

Creates a descriptive summary table with a single "Total" column format. By default (consider\_normality = "ROBUST"), continuous variables are shown as mean +/- SD or median [IQR] based on a four-gate decision ( $n < 3$  fail-safe, skewness/kurtosis, CLT, and Shapiro-Wilk). This can be overridden via consider\_normality and force\_ordinal.

## Usage

```
ternD(
  data,
  vars = NULL,
  exclude_vars = NULL,
  force_ordinal = NULL,
```

```

force_normal = NULL,
force_continuous = NULL,
output_xlsx = NULL,
output_docx = NULL,
consider_normality = "ROBUST",
print_normality = FALSE,
round_intg = FALSE,
round_decimal = NULL,
smart_rename = TRUE,
insert_subheads = TRUE,
factor_order = "mixed",
methods_doc = TRUE,
methods_filename = "TernTables_methods.docx",
category_start = NULL,
plain_header = NULL,
table_font_size = 9,
manual_italic_indent = NULL,
manual_underline = NULL,
table_caption = NULL,
table_footnote = NULL,
abbreviation_footnote = NULL,
variable_footnote = NULL,
index_style = "symbols",
line_break_header = getOption("TernTables.line_break_header", TRUE),
open_doc = TRUE,
citation = TRUE,
font_family = getOption("TernTables.font_family", "Arial"),
show_missing = FALSE,
zero_to_dash = FALSE,
show_missingness = FALSE,
missing_indicators = NULL
)

```

### Arguments

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>data</code>             | Tibble with variables.                                                                                                                                                                                                                                                                                                                                                                                            |
| <code>vars</code>             | Character vector of variables to summarize. Defaults to all except <code>exclude_vars</code> .                                                                                                                                                                                                                                                                                                                    |
| <code>exclude_vars</code>     | Character vector to exclude from the summary.                                                                                                                                                                                                                                                                                                                                                                     |
| <code>force_ordinal</code>    | Character vector of variables to treat as ordinal (i.e., use median [IQR]) regardless of the <code>consider_normality</code> setting. This parameter takes priority over normality testing when <code>consider_normality = "ROBUST"</code> or <code>TRUE</code> .                                                                                                                                                 |
| <code>force_normal</code>     | Character vector of variable names to treat as normally distributed, bypassing all normality assessment. Listed variables are summarized as mean $\pm$ SD regardless of the <code>consider_normality</code> setting. Takes priority over <code>consider_normality</code> but not over <code>force_ordinal</code> (if a variable appears in both, <code>force_ordinal</code> wins). Default is <code>NULL</code> . |
| <code>force_continuous</code> | Character vector of variables to force treatment as continuous (mean $\pm$ SD), bypassing the automatic binary 0/1 detection that would otherwise convert them to categorical Y/N. Useful when a numeric variable with only two unique values (e.g. 0/1 dose levels) should be analysed as a continuous measurement rather than a dichotomous category. Default is <code>NULL</code> .                            |

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| output_xlsx        | Optional Excel filename to export the table.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| output_docx        | Optional Word filename to export the table.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| consider_normality | Character or logical; controls routing of continuous variables to mean $\pm$ SD vs median [IQR]. "ROBUST" (default) applies a four-gate decision: (1) $n < 3 \rightarrow$ non-parametric (conservative fail-safe); (2) absolute skewness $> 2$ or excess kurtosis $> 7 \rightarrow$ non-parametric regardless of $n$ ; (3) $n \geq 30 \rightarrow$ parametric via the Central Limit Theorem; (4) otherwise Shapiro-Wilk $p > 0.05 \rightarrow$ parametric. If TRUE, uses Shapiro-Wilk alone (can be over-sensitive at large $n$ ). If FALSE, defaults to mean $\pm$ SD for all numeric variables unless specified in force_ordinal.                                                                       |
| print_normality    | Logical; if TRUE, includes Shapiro-Wilk P values as an additional column in the output. Default is FALSE.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| round_intg         | Logical; if TRUE, rounds all means, medians, IQRs, and standard deviations to nearest integer (0.5 rounds up). Default is FALSE.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| round_decimal      | Integer or NULL; number of decimal places for all continuous summary values (means, SDs, medians, IQRs). Overrides the default of 1 decimal place when set. Ignored when round_intg = TRUE. Default is NULL (1 decimal place).                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| smart_rename       | Logical; if TRUE, automatically cleans variable names and subheadings for publication-ready output using built-in rule-based pattern matching for common medical abbreviations and prefixes. Default is TRUE.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| insert_subheads    | Logical; if TRUE (default), creates a hierarchical structure with a header row and indented sub-category rows for categorical variables with 3 or more levels. Binary variables (Y/N, YES/NO, or numeric 1/0 – which are auto-detected and treated as Y/N) are always displayed as a single row showing the positive/yes count regardless of this setting. Two-level categorical variables whose values are not Y/N, YES/NO, or 1/0 (e.g. Male/Female) use the hierarchical sub-row format, showing both levels as indented rows. If FALSE, all categorical variables use a single-row flat format. Default is TRUE.                                                                                      |
| factor_order       | Character; controls the ordering of factor levels in the output. "mixed" (default) applies level-aware ordering for two-level categorical variables and frequency ordering for variables with three or more levels: for any factor, factor level order is always respected regardless of the number of levels; for non-factor two-level variables, levels are sorted alphabetically; for non-factor variables with three or more levels, levels are sorted by decreasing frequency. "levels" respects the original factor level ordering for all variables; if the variable is not a factor, falls back to frequency ordering. "frequency" orders all levels by decreasing frequency (most common first). |
| methods_doc        | Logical; if TRUE (default), generates a methods document describing the statistical presentation used. The document contains boilerplate text for all three table types so the relevant section can be copied directly into a manuscript.                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| methods_filename   | Character; filename for the methods document. Default is "TernTables_methods.docx".                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| category_start     | Named character vector specifying where to insert category headers. Names are the header label text to display; values are the anchor variable – either the original column name (e.g. "Age_Years") or the cleaned display name (e.g. "Age (yr)"). Both forms are accepted. Example: c("Demographics" = "Age_Years", "Clinical Measures" = "bmi"). Default is NULL (no category headers).                                                                                                                                                                                                                                                                                                                 |

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| plain_header          | Named character vector, same interface as <code>category_start</code> . Names are the label text; values are the anchor variable to insert before. Inserts a label-only row with underline formatting and no bold, merge, or border treatments. Default NULL.                                                                                                                                                                          |
| table_font_size       | Numeric; font size for Word document output tables. Default is 9.                                                                                                                                                                                                                                                                                                                                                                      |
| manual_italic_indent  | Character vector of display variable names (post-cleaning) that should be formatted as italicized and indented in Word output – matching the appearance of factor sub-category rows. Has no effect on the returned tibble; only applies when <code>output_docx</code> is specified. Default is NULL.                                                                                                                                   |
| manual_underline      | Character vector of display variable names (post-cleaning) that should be formatted as underlined in Word output – matching the appearance of multi-category variable headers. Has no effect on the returned tibble; only applies when <code>output_docx</code> is specified. Default is NULL.                                                                                                                                         |
| table_caption         | Optional character string for a table caption to display above the table in the Word document. Rendered as size 11 Arial bold, single-spaced with a small gap before the table. Default is NULL (no caption). Example: "Table 1. Patient demographics."                                                                                                                                                                                |
| table_footnote        | Optional character string for a footnote to display below the table in the Word document. Rendered as size 6 Arial italic with a double-bar border above and below. Default is NULL (no footnote).                                                                                                                                                                                                                                     |
| abbreviation_footnote | Optional character string listing abbreviations. Always printed first in the footnote block. Default NULL.                                                                                                                                                                                                                                                                                                                             |
| variable_footnote     | Optional named character vector. Names are display variable names (case-insensitive); values are the footnote definition text. Each variable gets the next symbol appended to its name in the table, and the footnote block lists each definition below the abbreviation line. To share one footnote between multiple variables, separate their names with a pipe: <code>c("Var A Var B" = "Shared note text.")</code> . Default NULL. |
| index_style           | Character; "symbols" (default) uses *, dagger, double-dagger ... "alphabet" uses Unicode superscript letters. See <code>word_export</code> for details.                                                                                                                                                                                                                                                                                |
| line_break_header     | Logical; if TRUE (default), column headers are wrapped with <code>\n</code> – the first column header includes a category hierarchy label, and the sample size appears on a second line. Set to FALSE to suppress all header line breaks. Can also be set package-wide via <code>options(TernTables.line_break_header = FALSE)</code> .                                                                                                |
| open_doc              | Logical; if TRUE (default), automatically opens the written Word document in the system default application after saving. Set to FALSE to suppress. Has no effect when <code>output_docx</code> is NULL.                                                                                                                                                                                                                               |
| citation              | Logical; if TRUE (default), appends a citation line at the bottom of the table footnote block and at the end of the methods document: package version, authors, and links to the GitHub repository and web interface. Set to FALSE to suppress.                                                                                                                                                                                        |
| font_family           | Character; font family name used for all Word output (table, captions, footnotes, methods document). Any font installed on the system that renders the document may be used. Popular options include "Arial", "Helvetica", "Times New Roman", "Garamond", and "Calibri". Defaults to <code>getOption("TernTables.font_family", "Arial")</code> .                                                                                       |

|                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>show_missing</code>       | Logical; if TRUE, appends a "Missing" row after each variable's data rows showing the count and percentage of missing observations (denominator is the total N). Only emitted when at least one observation is missing. A footnote is automatically appended noting that missing values are reported. Default is FALSE.                                                                                                                                                                                                                                                                                                                        |
| <code>zero_to_dash</code>       | Logical; if TRUE, replaces any categorical cell that would display "0 (0%)" with "-" in the output table. Useful when zero counts are not meaningful to report numerically. Default is FALSE.                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <code>show_missingness</code>   | Controls whether a "Missing, n (%)" column is appended to the table after the Total column. Options:<br>FALSE (default) — no missingness column added.<br>"total" — one column appended showing the count and percentage of missing observations across all rows for each variable.<br>"group" is not supported by <code>ternD()</code> (which has no group structure); use <code>ternG()</code> with <code>show_missingness = "group"</code> instead. Missingness is computed on the raw data column so both NA values and string representations of missing data (e.g., "Unknown", "N/A") are counted. See <code>missing_indicators</code> . |
| <code>missing_indicators</code> | Optional character vector of string values to treat as missing in addition to (or instead of) the built-in <code>ternP</code> defaults. When NULL (default), the <code>ternP</code> canonical list is used. When supplied, the custom list <b>replaces</b> the <code>ternP</code> defaults. Matching is case-insensitive and trims whitespace.                                                                                                                                                                                                                                                                                                 |

## Details

The function always returns a tibble with a single Total (N = n) column format, regardless of the `consider_normality` setting. The behavior for numeric variables follows this priority:

1. Variables in `force_ordinal`: Always use median [IQR]
2. When `consider_normality = "ROBUST"`: Four-gate decision (n<3 fail-safe, skewness/kurtosis, CLT, Shapiro-Wilk)
3. When `consider_normality = TRUE`: Use Shapiro-Wilk test to choose format
4. When `consider_normality = FALSE`: Default to mean +/- SD

For categorical variables, the function shows frequencies and percentages. When `insert_subheads = TRUE`, categorical variables with 3 or more levels are displayed with hierarchical formatting (main variable as header, levels as indented sub-rows). Binary variables (Y/N, YES/NO, or numeric 1/0 auto-detected as Y/N) always use a single-row format showing only the positive/yes count, regardless of this setting. Two-level categorical variables whose values are not Y/N, YES/NO, or 1/0 (e.g. Male/Female) also use the hierarchical sub-row format.

## Value

A tibble with one row per variable (multi-row for factors), containing:

**Variable** Variable names with appropriate indentation

**Total (N = n)** Summary statistics (mean +/- SD, median [IQR], or n (%) as appropriate)

**SW\_p** Shapiro-Wilk P values (only if `print_normality = TRUE`)

## Examples

```
data(tern_colon)

# Basic descriptive summary
ternD(tern_colon, exclude_vars = c("ID"), methods_doc = FALSE)

# With normality-aware formatting and category section headers
ternD(tern_colon, exclude_vars = c("ID"), methods_doc = FALSE,
      category_start = c("Patient Demographics" = "Age (yr)",
                        "Tumor Characteristics" = "Positive Lymph Nodes (n)"))

# Force specific variables to ordinal (median [IQR]) display
ternD(tern_colon, exclude_vars = c("ID"), methods_doc = FALSE,
      force_ordinal = c("Positive_Lymph_Nodes_n"))

# Export to Word (writes a file to tempdir)

ternD(tern_colon,
      exclude_vars = c("ID"),
      methods_doc = FALSE,
      open_doc = FALSE,
      output_docx = file.path(tempdir(), "descriptive.docx"),
      category_start = c("Patient Demographics" = "Age (yr)",
                        "Surgical Findings" = "Colonic Obstruction",
                        "Tumor Characteristics" = "Positive Lymph Nodes (n)",
                        "Outcomes" = "Recurrence"))
```

---

|       |                                                                          |
|-------|--------------------------------------------------------------------------|
| ternG | <i>Generate grouped summary table with appropriate statistical tests</i> |
|-------|--------------------------------------------------------------------------|

---

## Description

Creates a grouped summary table with optional statistical testing for group comparisons. Supports numeric and categorical variables; numeric variables can be treated as ordinal via `force_ordinal`. Includes options to calculate P values and odds ratios. For descriptive (ungrouped) tables, use `ternD`.

## Usage

```
ternG(
  data,
  vars = NULL,
  exclude_vars = NULL,
  group_var,
  force_ordinal = NULL,
  force_normal = NULL,
  force_continuous = NULL,
  group_order = NULL,
  output_xlsx = NULL,
  output_docx = NULL,
  OR_col = FALSE,
  OR_method = "dynamic",
```

```

consider_normality = "ROBUST",
print_normality = FALSE,
show_test = FALSE,
p_digits = 3,
round_intg = FALSE,
round_decimal = NULL,
smart_rename = TRUE,
insert_subheads = TRUE,
factor_order = "mixed",
table_font_size = 9,
methods_doc = TRUE,
methods_filename = "TernTables_methods.docx",
category_start = NULL,
plain_header = NULL,
manual_italic_indent = NULL,
manual_underline = NULL,
indent_info_column = FALSE,
show_total = TRUE,
table_caption = NULL,
table_footnote = NULL,
abbreviation_footnote = NULL,
variable_footnote = NULL,
index_style = "symbols",
line_break_header = getOption("TernTables.line_break_header", TRUE),
post_hoc = FALSE,
p_adjust = FALSE,
p_adjust_display = "fdr_only",
open_doc = TRUE,
citation = TRUE,
font_family = getOption("TernTables.font_family", "Arial"),
show_missing = FALSE,
show_p = TRUE,
zero_to_dash = FALSE,
percentage_compute = "column",
categorical_posthoc = FALSE,
show_missingness = FALSE,
missing_indicators = NULL
)

```

### Arguments

|                            |                                                                                                                                                          |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>data</code>          | Tibble containing all variables.                                                                                                                         |
| <code>vars</code>          | Character vector of variables to summarize. Defaults to all except <code>group_var</code> and <code>exclude_vars</code> .                                |
| <code>exclude_vars</code>  | Character vector of variable(s) to exclude. <code>group_var</code> is automatically excluded.                                                            |
| <code>group_var</code>     | Character, the grouping variable (factor or character with $\geq 2$ levels).                                                                             |
| <code>force_ordinal</code> | Character vector of variables to treat as ordinal (i.e., use medians/IQR and non-parametric tests).                                                      |
| <code>force_normal</code>  | Character vector of variable names to treat as normally distributed, bypassing all normality assessment (Gates 1–4 under "ROBUST", or Shapiro-Wilk under |

TRUE). Listed variables are summarized as mean  $\pm$  SD and compared with Welch tests regardless of the `consider_normality` setting. Takes priority over `consider_normality` but not over `force_ordinal` (if a variable appears in both, `force_ordinal` wins). Default is NULL.

#### `force_continuous`

Character vector of variables to force treatment as continuous (mean  $\pm$  SD and parametric tests), bypassing the automatic binary 0/1 detection that would otherwise convert them to categorical Y/N. Useful when a numeric variable with only two unique values (e.g. 0/1 dose levels) should be analysed as a continuous measurement rather than a dichotomous category. Takes priority over automatic type detection but does not override `force_ordinal` (if a variable appears in both, `force_ordinal` wins). Default is NULL.

`group_order` Optional character vector to specify a custom group level order.

`output_xlsx` Optional filename to export the table as an Excel file.

`output_docx` Optional filename to export the table as a Word document.

`OR_col` Logical; if TRUE, adds unadjusted odds ratios with 95% CI for binary categorical variables (Y/N, YES/NO, or numeric 0/1) and two-level categorical variables (e.g. Male/Female). For two-level categoricals displayed with sub-rows, the reference level (factor level 1, or alphabetical first for non-factors) shows "1.00 (ref.)"; the non-reference level shows the computed OR with 95% CI. Variables with three or more levels show "-". Only valid when `group_var` has exactly 2 levels; an error is raised for 3+ group comparisons. Default is FALSE.

`OR_method` Character; controls how odds ratios are calculated when `OR_col` = TRUE. If "dynamic" (default), uses Fisher's exact method when any expected cell count is  $< 5$  (Cochran criterion), otherwise uses the Wald method. If "wald", forces the Wald method regardless of expected cell counts.

#### `consider_normality`

Character or logical; controls how continuous variables are routed to parametric vs. non-parametric tests. "ROBUST" (default) applies a four-gate decision consistent with standard biostatistical practice: (1) any group  $n < 3$  is a conservative fail-safe to non-parametric; (2) absolute skewness  $> 2$  or excess kurtosis  $> 7$  in any group routes to non-parametric regardless of sample size (catches heavy-tailed distributions and LOS/count-style skews in which the CLT guarantee is compromised); (3) all groups  $n \geq 30$  routes to parametric via the Central Limit Theorem; (4) otherwise Shapiro-Wilk  $p > 0.05$  in all groups routes to parametric. Normal variables use mean  $\pm$  SD and Welch t-test (2 groups) or Welch ANOVA (3+ groups); non-normal variables use median [IQR] and Wilcoxon rank-sum (2 groups) or Kruskal-Wallis (3+ groups). If TRUE, uses Shapiro-Wilk alone ( $p > 0.05$  in all groups = normal). Conservative at large  $n$ . If FALSE, all numeric variables are treated as normally distributed regardless of distribution. If "FORCE", all numeric variables are treated as non-normal (median [IQR], nonparametric tests).

#### `print_normality`

Logical; if TRUE, includes Shapiro-Wilk P values in the output. Default is FALSE.

`show_test` Logical; if TRUE, includes the statistical test name as a column in the output. Default is FALSE.

`p_digits` Integer; number of decimal places for P values (default 3).

`round_intg` Logical; if TRUE, rounds all means, medians, IQRs, and standard deviations to nearest integer (0.5 rounds up). Default is FALSE.

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| round_decimal        | Integer or NULL; number of decimal places for all continuous summary values (means, SDs, medians, IQRs). Overrides the default of 1 decimal place when set. Ignored when round_intg = TRUE. Default is NULL (1 decimal place).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| smart_rename         | Logical; if TRUE, automatically cleans variable names and subheadings for publication-ready output using built-in rule-based pattern matching for common medical abbreviations and prefixes. Default is TRUE.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| insert_subheads      | Logical; if TRUE (default), creates a hierarchical structure with a header row and indented sub-category rows for categorical variables with 3 or more levels. Binary variables (Y/N, YES/NO, or numeric 1/0 – which are auto-detected and treated as Y/N) are always displayed as a single row showing the positive/yes count regardless of this setting. Two-level categorical variables whose values are not Y/N, YES/NO, or 1/0 (e.g. Male/Female) use the hierarchical sub-row format, showing both levels as indented rows. If FALSE, all categorical variables use a single-row flat format. Default is TRUE.                                                                                                         |
| factor_order         | Character; controls the ordering of factor levels in the output. "mixed" (default) applies level-aware ordering for two-level categorical variables and frequency ordering for variables with three or more levels: for any factor, factor level order is always respected regardless of the number of levels; for non-factor two-level variables (e.g. Male/Female), levels are sorted alphabetically; for non-factor variables with three or more levels, levels are sorted by decreasing frequency. "levels" respects the original factor level ordering for all variables; if the variable is not a factor, falls back to frequency ordering. "frequency" orders all levels by decreasing frequency (most common first). |
| table_font_size      | Numeric; font size for Word document output tables. Default is 9.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| methods_doc          | Logical; if TRUE (default), generates a methods document describing the statistical tests used.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| methods_filename     | Character; filename for the methods document. Default is "TernTables_methods.docx".                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| category_start       | Named character vector specifying where to insert category headers. Names are the header label text to display; values are the anchor variable – either the original column name (e.g. "Age_Years") or the cleaned display name (e.g. "Age (yr)"). Both forms are accepted. Example: c("Demographics" = "Age_Years", "Clinical" = "bmi"). Default is NULL (no category headers).                                                                                                                                                                                                                                                                                                                                             |
| plain_header         | Named character vector, same interface as category_start. Names are the label text; values are the anchor variable to insert before. Inserts a label-only row with underline formatting and no bold, merge, or border treatments. Default NULL.                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| manual_italic_indent | Character vector of display variable names (post-cleaning) that should be formatted as italicized and indented in Word output – matching the appearance of factor sub-category rows. Has no effect on the returned tibble; only applies when output_docx is specified or when the tibble is passed to word_export.                                                                                                                                                                                                                                                                                                                                                                                                           |
| manual_underline     | Character vector of display variable names (post-cleaning) that should be formatted as underlined in Word output – matching the appearance of multi-category variable headers. Has no effect on the returned tibble; only applies when output_docx is specified or when the tibble is passed to word_export.                                                                                                                                                                                                                                                                                                                                                                                                                 |

|                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>indent_info_column</code>    | Logical; if FALSE (default), the internal <code>.indent</code> helper column is dropped from the returned tibble. Set to TRUE to retain it – this is necessary when you intend to post-process the tibble and later pass it to <code>word_export</code> directly, as <code>word_export</code> uses the <code>.indent</code> column to apply correct indentation and italic formatting in the Word table.                                                                                                                                                                                                                                                                                                                            |
| <code>show_total</code>            | Logical; if TRUE, adds a "Total" column showing the aggregate summary statistic across all groups (e.g., for a publication Table 1 that includes both per-group and overall columns). Default is TRUE.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <code>table_caption</code>         | Optional character string for a table caption to display above the table in the Word document. Rendered as size 11 Arial bold, single-spaced with a small gap before the table. Default is NULL (no caption). Example: "Table 2. Comparison of recurrence vs. no recurrence."                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <code>table_footnote</code>        | Optional character string for a footnote to display below the table in the Word document. Rendered as size 6 Arial italic with a double-bar border above and below. Default is NULL (no footnote).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <code>abbreviation_footnote</code> | Optional character string listing abbreviations. Always printed first in the footnote block. Default NULL.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <code>variable_footnote</code>     | Optional named character vector. Names are display variable names (case-insensitive); values are the footnote definition text. Each variable gets the next symbol appended to its name in the table, and the footnote block lists each definition below the abbreviation line. To share one footnote between multiple variables, separate their names with a pipe: <code>c("Var A Var B" = "Shared note text.")</code> . Default NULL.                                                                                                                                                                                                                                                                                              |
| <code>index_style</code>           | Character; "symbols" (default) uses *, dagger, double-dagger ... "alphabet" uses Unicode superscript letters. See <code>word_export</code> for details.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <code>line_break_header</code>     | Logical; if TRUE (default), column headers are wrapped with <code>\n</code> – group names break on spaces, sample size counts move to a second line, and the first column header reads "Category / Variable". Set to FALSE to suppress all header line breaks. Can also be set package-wide via <code>options(TernTables.line_break_header = FALSE)</code> .                                                                                                                                                                                                                                                                                                                                                                        |
| <code>post_hoc</code>              | Logical; if TRUE, runs pairwise post-hoc tests for continuous and ordinal variables in three or more group comparisons and annotates each group column value with a compact letter display (CLD) superscript. Groups sharing a letter are not significantly different at $\alpha = 0.05$ . For normally distributed variables (Welch ANOVA path), Games-Howell pairwise tests are used. For non-normal and ordinal variables (Kruskal-Wallis path), Dunn's test with Holm correction is used. Post-hoc testing is never applied to categorical variables. Only valid when <code>group_var</code> has three or more levels; silently ignored for two-group comparisons. Requires the <code>rstatix</code> package. Default is FALSE. |
| <code>p_adjust</code>              | Logical; if TRUE, applies the Benjamini-Hochberg (BH) false discovery rate correction to all omnibus P values after testing. The correction pool is one P value per variable — sub-rows of multi-level categoricals share the parent P value and are not double-counted. Post-hoc pairwise P values (which already carry within-variable Holm correction) are excluded from the correction pool. Default is FALSE.                                                                                                                                                                                                                                                                                                                  |

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| p_adjust_display    | Character; controls how BH-corrected P values appear in the output when p_adjust = TRUE. "fdr_only" (default) replaces the standard P value column with the corrected values, renaming the column to "P value (FDR corrected)". "both" retains the original P values in a column named "P value" and adds FDR-corrected values immediately to its right in a column named "P value (FDR corrected)". Ignored when p_adjust = FALSE.                                                                                                                                                                                                                                                                                                                                                               |
| open_doc            | Logical; if TRUE (default), automatically opens the written Word document in the system default application after saving. Set to FALSE to suppress. Has no effect when output_docx is NULL.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| citation            | Logical; if TRUE (default), appends a citation line at the bottom of the table footnote block and at the end of the methods document: package version, authors, and links to the GitHub repository and web interface. Set to FALSE to suppress.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| font_family         | Character; font family name used for all Word output (table, captions, footnotes, methods document). Any font installed on the system that renders the document may be used. Popular options include "Arial", "Helvetica", "Times New Roman", "Garamond", and "Calibri". Defaults to getOption("TernTables.font_family", "Arial").                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| show_missing        | Logical; if TRUE, appends a "Missing" row after each variable's data rows showing the count and percentage of missing observations per group (denominator is each group's total N). Missing rows display "-" in the P and OR columns. A footnote is automatically appended noting that missing values are reported. Default is FALSE.                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| show_p              | Logical; if TRUE (default), the P value column is included in the output and Excel/Word exports. Set to FALSE to produce a descriptive-only grouped table — the output will contain only the Variable column, one column per group level, and the Total column (if show_total = TRUE). When FALSE, OR_col, show_test, print_normality, post_hoc, categorical_posthoc, and p_adjust are all suppressed automatically.                                                                                                                                                                                                                                                                                                                                                                              |
| zero_to_dash        | Logical; if TRUE, replaces any categorical cell that would display "0 (0%)" with "-" in the output table. Useful when zero counts in a group are not meaningful to report numerically (e.g. no patients with a condition in one arm). Default is FALSE.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| percentage_compute  | Character; controls the denominator used when computing percentages for categorical variables. "column" (default) divides each cell count by the column (group) total, so percentages describe the composition of each group — the standard Table 1 interpretation (e.g. "60% of the Recurrence group is Male"). "row" divides each cell count by the row total (the number of subjects with that category level across all groups), so percentages describe how each category level is distributed across groups (e.g. "30% of Males had Recurrence"). When "row", the Total column is automatically suppressed (a Total column would show 100% for every level, which is uninformative). Applies to both binary and multinomial categorical variables in both two- and three-group comparisons. |
| categorical_posthoc | Logical; if TRUE, computes adjusted standardized residuals from the global contingency table for categorical variables following a significant omnibus test ( $p < 0.05$ ). Cells whose adjusted standardized residual exceeds $\pm 1.96$ are marked with an asterisk (*), indicating a significant deviation from expected frequencies ( $\alpha = 0.05$ ). This method is equivalent to Haberman's adjusted residuals and does not require a separate multiple-comparisons correction, as the $\pm 1.96$                                                                                                                                                                                                                                                                                        |

threshold is derived directly from the omnibus test distribution. Only applied when `group_var` has three or more levels; silently ignored for two-group comparisons. Default is `FALSE`.

#### `show_missingness`

Controls whether a column of missing-value percentages is appended to the table. Options:

`FALSE` (default) — no missingness columns added.

`"total"` — one column (`"Missing, n (%)"`) appended at the far right showing the number and percentage of missing observations across *all* rows for each variable.

`"group"` — one column per group level (`"Miss. [level]"`) interleaved immediately after each group's data column, showing per-group missingness for each variable.

Missingness is computed on the *raw* data column (before ternG's internal NA filtering), so both NA values and string representations of missing data (e.g., `"Unknown"`, `"N/A"`) are counted. See `missing_indicators` to customise which strings count.

#### `missing_indicators`

Optional character vector of string values to treat as missing in addition to (or instead of) the built-in ternP defaults. When `NULL` (default), the ternP canonical list is used (`"na"`, `"n/a"`, `"unknown"`, etc.). When supplied, the custom list **replaces** the ternP defaults — only the values in `missing_indicators` (plus true NA) are counted as missing. Matching is always case-insensitive and ignores leading/trailing whitespace.

## Details

**Independence assumption:** all statistical tests applied by this function (Welch's *t*-test, Wilcoxon rank-sum, Welch ANOVA, Kruskal-Wallis, chi-squared, and Fisher's exact) assume that observations are independent — each row must represent a distinct, unrelated subject. ternG is not appropriate for repeated-measures, longitudinal, or clustered data (e.g. pre/post measurements, matched pairs, or patients nested within sites).

## Value

A tibble with one row per variable (multi-row for multi-level factors), showing summary statistics by group, P values, test type, and optionally odds ratios and total summary column.

## Examples

```
data(tern_colon)

# 2-group comparison
ternG(tern_colon, exclude_vars = c("ID"), group_var = "Recurrence",
      methods_doc = FALSE)

# 2-group comparison with odds ratios
ternG(tern_colon, exclude_vars = c("ID"), group_var = "Recurrence",
      OR_col = TRUE, methods_doc = FALSE)

# 3-group comparison
ternG(tern_colon, exclude_vars = c("ID"), group_var = "Treatment_Arm",
      group_order = c("Observation", "Levamisole", "Levamisole + 5FU"),
      methods_doc = FALSE)
```

```
# 2-group comparison with BH FDR correction (fdr_only – default display)
ternG(tern_colon, exclude_vars = c("ID"), group_var = "Recurrence",
      p_adjust = TRUE, methods_doc = FALSE)

# 2-group comparison with BH FDR correction (show raw + corrected side by side)
ternG(tern_colon, exclude_vars = c("ID"), group_var = "Recurrence",
      p_adjust = TRUE, p_adjust_display = "both", methods_doc = FALSE)

# Export to Word (writes a file to tempdir)

ternG(tern_colon,
      exclude_vars = c("ID"),
      group_var = "Recurrence",
      OR_col = TRUE,
      methods_doc = FALSE,
      open_doc = FALSE,
      output_docx = file.path(tempdir(), "comparison.docx"),
      category_start = c("Patient Demographics" = "Age (yr)",
                        "Tumor Characteristics" = "Positive Lymph Nodes (n)"))
```

---

ternP

---

*Preprocess a raw data frame for use with ternG or ternD*


---

## Description

ternP() cleans a raw data frame loaded from a CSV or XLSX file, applying a standardized set of transformations and performing validation checks before the data is passed to [ternG](#) or [ternD](#).

## Usage

```
ternP(data, mode = "auto", extra_na = NULL, drop_cols = NULL)
```

## Arguments

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| data | A data frame or tibble as loaded from a CSV or XLSX file (e.g. via <code>readr::read_csv()</code> or <code>readxl::read_excel()</code> ). All character columns are processed; numeric and logical columns are passed through unchanged by the string-cleaning steps.                                                                                                                                                                                                                                                                                                                                                                                    |
| mode | Preprocessing mode. One of "auto" (default) or "manual". <p>"auto" Default behaviour. PHI column-name detection and unnamed-column checks run as hard stops before any cleaning. All other transformations (string-NA conversion, whitespace trimming, empty-column removal, blank-row removal, case normalisation) run automatically.</p> <p>"manual" PHI detection is <b>skipped</b>. You take full responsibility for ensuring no patient identifiers are present. A prominent warning is emitted. All other cleaning transformations still run. Use <code>drop_cols</code> to explicitly remove any columns you do not want in the cleaned data.</p> |

|           |                                                                                                                                                                                                                                                                                                                                                        |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| extra_na  | Optional character vector of additional string values to treat as missing (converted to NA). These are <b>appended</b> to the built-in list (na, n/a, missing, unknown, etc.) — not a replacement. Matching is case-insensitive and whitespace-trimmed. Works in both "auto" and "manual" modes. Example: extra_na = c("9999", "Not Done", "PENDING"). |
| drop_cols | Optional character vector of column names to drop from the data before cleaning begins. Intended for use in "manual" mode to explicitly remove identifier or unwanted columns without triggering the PHI check. Any names not found in data are silently ignored. Works in both modes.                                                                 |

## Value

A named list with three elements:

**clean\_data** A tibble containing the fully cleaned dataset, ready to pass to ternG() or ternD().

**sparse\_rows** A tibble of rows from clean\_data where more than 50% of values are NA. These rows are *retained* in clean\_data but extracted here for optional review or download. An empty tibble if no sparse rows exist.

**feedback** A named list of feedback items. Each element is NULL if the corresponding transformation was not triggered, or a value describing what changed:

**string\_na\_converted** A named list with elements total (integer count of values converted) and cols (character vector of affected column names), or NULL if no string NA values were found.

**blank\_rows\_removed** A named list with elements count (integer) and row\_indices (integer vector of original row positions removed), or NULL if none.

**sparse\_rows\_flagged** A named list with elements count (integer) and row\_indices (integer vector of row positions in clean\_data with >50% missingness), or NULL if none.

**case\_normalized\_vars** A named list with elements cols (character vector of affected column names) and detail (a named list per column, each with changed\_from and changed\_to character vectors showing the exact value changes), or NULL if none. #

**dropped\_user\_cols** Character vector of column names explicitly dropped via the drop\_cols parameter, or NULL if drop\_cols was not used.

**manual\_mode** Logical. TRUE when mode = "manual" was used (PHI check skipped), FALSE otherwise.

**dropped\_empty\_cols** Character vector of column names (or "" for unnamed columns) that were dropped because they were 100% empty, or NULL if none.

**date\_cols\_detected** Character vector of column names that appear to contain date values — either R Date/POSIXct types (from Excel) or character columns where ≥80% of non-NA values match a common date pattern (from CSV). These columns are *not* dropped automatically; the caller should decide whether to exclude them or keep them as categorical variables.

## Cleaning pipeline (in order)

1. Date columns are detected (R Date/POSIXct types, or character columns where ≥80% of values match a common date pattern) and reported in feedback\$date\_cols\_detected. They are not dropped automatically — the caller decides whether to exclude or keep them.
2. String NA values ("NA", "na", "N/A", "NaN", "missing", "unknown", "unk", "not available", "not applicable", "none", "null", "nil", "-", ".", "?") are converted to NA (matching is case-insensitive).
3. Leading and trailing whitespace is trimmed from all character columns.

4. Columns that are 100% empty (all NA) are silently dropped.
5. Rows where every cell is NA are removed.
6. Character columns where values differ only by capitalization (e.g. "Male" vs "MAle") are standardized to title case.

### Validation hard stops

ternP() stops with a descriptive error if:

- Any column name matches a protected health information (PHI) pattern (e.g. MRN, DOB, FirstName). De-identified research identifiers such as patient\_id, subject\_id, and participant\_id are explicitly excluded, as are clinical-event dates (admission date, discharge date, visit date, etc.). Only personal-identity dates such as DOB and DOD are flagged.
- Any column with a blank or whitespace-only header contains data. Completely empty unnamed columns are silently dropped and do not trigger this error.

### See Also

[ternG](#) for grouped comparisons, [ternD](#) for descriptive statistics.

### Examples

```
# Load a messy CSV and preprocess it
path <- system.file("extdata/csv", "tern_colon_messy.csv",
                    package = "TernTables")
raw <- read.csv(path, stringsAsFactors = FALSE)
result <- ternP(raw)

# Access cleaned data
result$clean_data

# Review preprocessing feedback
result$feedback

# Sparse rows flagged (>50% missing), retained but not removed
result$sparse_rows
```

---

ternStyle

*Export a custom tibble to Word with TernTables formatting*

---

### Description

ternStyle() renders any user-built tibble into a Word document with the exact same visual style as tables produced by ternG(), ternD(), and word\_export() – Arial font, grey header, double-bar footer, caption/footnote block, and citation footer.

**Usage**

```

ternStyle(
  tbl,
  filename = NULL,
  col1_name = NULL,
  subheader_rows = NULL,
  bold_rows = NULL,
  bold_sig = NULL,
  italic_rows = NULL,
  bold_cols = NULL,
  italic_cols = NULL,
  header_format_follow = FALSE,
  round_intg = FALSE,
  round_decimal = NULL,
  font_size = 9,
  category_start = NULL,
  plain_header = NULL,
  manual_italic_indent = NULL,
  manual_underline = NULL,
  table_caption = NULL,
  table_footnote = NULL,
  abbreviation_footnote = NULL,
  variable_footnote = NULL,
  index_style = "symbols",
  col1_header = NULL,
  line_break_header = FALSE,
  open_doc = TRUE,
  citation = TRUE,
  font_family = getOption("TernTables.font_family", "Arial")
)

```

**Arguments**

|                             |                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>tbl</code>            | A data frame or tibble. The first column is used as the row-label column (rendered as "Variable" unless renamed via <code>col1_name</code> ). All columns are coerced to character before rendering; NA values become empty strings.                                                                                                                                                   |
| <code>filename</code>       | Output file path ending in .docx. Pass NULL (default) to write to a temporary file and suppress auto-opening – useful when the result will be passed directly to <a href="#">ternB</a> for bundling.                                                                                                                                                                                   |
| <code>col1_name</code>      | Optional character string. If supplied, the first column is renamed to this label in the rendered table. The column need not be named "Variable" in the input; any name is accepted and renamed here. Default NULL (use the tibble's existing first column name).                                                                                                                      |
| <code>subheader_rows</code> | Character vector of labels that already exist as rows in <code>tbl</code> and should be formatted as full section-header rows: cells merged across all columns, bold, with a bottom border line – identical to the treatment applied by <code>category_start</code> . No row is inserted; the matching existing row is formatted in place. Matching is case-insensitive. Default NULL. |
| <code>bold_rows</code>      | Integer vector of body row indices (1-based, final rendered table) to bold across every column. Applied after all structural formatting so it always wins. Default NULL.                                                                                                                                                                                                               |

|                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>bold_sig</b>              | <p>Optional named list for cell-level p-value-based bolding. Use this when your tibble has pre-formatted p-value strings in columns that are not named "P value" (e.g. "Uni p", "Multi p"). Supply a list with:</p> <ul style="list-style-type: none"> <li>• <b>p_cols</b> — character vector of column names containing p-value strings.</li> <li>• <b>hr_cols</b> — optional character vector of column names (same length and order as <b>p_cols</b>) to also bold when the paired p-value is significant (e.g. the corresponding HR or coefficient column). Pass <b>NULL</b> or omit to skip paired-column bolding.</li> <li>• <b>threshold</b> — numeric significance threshold. Default 0.05.</li> </ul> <p>The <b>Variable</b> column is never modified by <b>bold_sig</b>; use <b>bold_rows</b> to bold entire rows (e.g., predictor-level rows where the p-value represents an omnibus LRT). Example:</p> <pre>bold_sig = list(   p_cols    = c("Uni p", "Multi p"),   hr_cols    = c("Uni HR (95% CI)", "Multi HR (95% CI)"),   threshold = 0.05 )</pre> <p>Default <b>NULL</b>.</p> |
| <b>italic_rows</b>           | Integer vector of body row indices to italicize across every column. Default <b>NULL</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>bold_cols</b>             | Integer vector of column indices (1-based) to bold across all body rows. Default <b>NULL</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>italic_cols</b>           | Integer vector of column indices to italicize across all body rows. Default <b>NULL</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>header_format_follow</b>  | Logical; if <b>TRUE</b> , columns listed in <b>bold_cols</b> or <b>italic_cols</b> also have their header cell bolded or italicized. Default <b>FALSE</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>round_intg</b>            | Logical; passed to <b>word_export</b> . Default <b>FALSE</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>round_decimal</b>         | Integer or <b>NULL</b> ; if provided, rounds all numeric values in the table to this many decimal places before rendering. Passed to <b>word_export</b> . Default <b>NULL</b> (no rounding).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>font_size</b>             | Numeric; font size for table body. Default 9.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>category_start</b>        | Named character vector; same as in <b>word_export</b> . Insert new section-header rows at anchor variable positions, in addition to any rows already in the tibble. Default <b>NULL</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>plain_header</b>          | Named character vector; same as in <b>word_export</b> . Insert underline-only (no bold, no merge) label rows at anchor positions. Default <b>NULL</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>manual_italic_indent</b>  | Character vector of row labels to italicize and indent (sub-item appearance). Default <b>NULL</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| <b>manual_underline</b>      | Character vector of row labels to underline (multi- category header appearance without the full subheader treatment). Default <b>NULL</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>table_caption</b>         | Optional character string for the caption above the table. Default <b>NULL</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>table_footnote</b>        | Optional character string for a footnote below the table. Default <b>NULL</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>abbreviation_footnote</b> | Optional character string (or character vector) of abbreviations. Always printed first in the footnote block. Default <b>NULL</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|                   |                                                                                                                                                                                                                                                                                                                |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| variable_footnote | Optional named character vector of per-variable footnote definitions (case-insensitive name match). To share one footnote symbol between multiple variables, separate their names with a pipe: <code>c("Var A Var B" = "Shared note text.")</code> . Default NULL.                                             |
| index_style       | Character; "symbols" (default) or "alphabet". Controls the footnote symbol sequence. See <code>word_export</code> for details.                                                                                                                                                                                 |
| col1_header       | Optional character string. Overrides the top-left header cell. When NULL (default), the standard "Category\n Variable" label is used. Example: "Variable\n Index Management Strategy".                                                                                                                         |
| line_break_header | Logical; if TRUE, column headers are wrapped with <code>\n</code> and the first column header shows the two-line "Category / Variable" label. For custom tibbles the column names are typically already formatted, so this defaults to FALSE here (unlike <code>word_export</code> where it defaults to TRUE). |
| open_doc          | Logical; if TRUE (default), opens the written document after saving. Default TRUE.                                                                                                                                                                                                                             |
| citation          | Logical; if TRUE (default), appends the TernTables citation line in the page footer. Default TRUE.                                                                                                                                                                                                             |
| font_family       | Character; font family name used for all Word output. Defaults to <code>getOption("TernTables.font_family", "Arial")</code> . See <code>word_export</code> for details.                                                                                                                                        |

## Details

Use this function when you have pre-computed summary statistics in a tibble (e.g. a custom cross-tab or manually assembled output table) and want it to match the rest of your TernTables document without running it through the full `ternG/ternD` pipeline.

## Value

Invisibly returns the input tibble (after renaming and coercion) with a "ternB\_meta" attribute attached. This makes the result directly passable to `ternB` for bundling with other tables into a combined Word document.

## Examples

```
library(tibble)
my_tbl <- tibble(
  Variable = c("Section A", "Row 1", "Row 2", "Section B", "Row 3"),
  `Group 1` = c("", "12 (40%)", "18 (60%)", "", "9 (30%)",
  `Group 2` = c("", "15 (50%)", "15 (50%)", "", "21 (70%)")
)
ternStyle(
  tbl = my_tbl,
  filename = file.path(tempdir(), "custom_table.docx"),
  subheader_rows = c("Section A", "Section B"),
  open_doc = FALSE,
  citation = FALSE
)
```

---

tern\_colon

*Colon Cancer Recurrence Data (Example Dataset)*


---

## Description

A processed subset of the [colon](#) dataset restricted to the recurrence endpoint (etype == 1), providing one row per patient. Variables have been relabelled with clinically descriptive names and factor levels suitable for direct use in TernTables functions. This dataset is provided as a ready-to-use example for demonstrating ternD() and ternG() functionality.

## Usage

```
tern_colon
```

## Format

A tibble with 929 rows and 12 variables:

**ID** Integer patient identifier.

**Age\_Years** Age at study entry (years).

**Sex** Patient sex: "Female" or "Male".

**Colonic\_Obstruction** Colonic obstruction present: "N" or "Y".

**Bowel\_Perforation** Bowel perforation present: "N" or "Y".

**Positive\_Lymph\_Nodes\_n** Number of positive lymph nodes detected.

**Over\_4\_Positive\_Nodes** More than 4 positive lymph nodes: "N" or "Y".

**Tumor\_Adherence** Tumour adherence to surrounding organs: "N" or "Y".

**Tumor\_Differentiation** Tumour differentiation grade: "Well", "Moderate", or "Poor".

**Extent\_of\_Local\_Spread** Depth of tumour penetration: "Submucosa", "Muscle", "Serosa", or "Contiguous Structures".

**Recurrence** Recurrence status: "No Recurrence" or "Recurrence".

**Treatment\_Arm** Randomised treatment: "Levamisole + 5FU", "Levamisole", or "Observation".

## Source

Derived from [colon](#) (Laurie et al., 1989). See [colon](#) for full provenance. Pre-processing script: data-raw/tern\_colon.R.

## Examples

```
data(tern_colon)
head(tern_colon)
```

---

|            |                                    |
|------------|------------------------------------|
| val_format | <i>Format a mean +/- SD string</i> |
|------------|------------------------------------|

---

**Description**

Format a mean +/- SD string

**Usage**

```
val_format(mean, sd)
```

**Arguments**

|      |                                                           |
|------|-----------------------------------------------------------|
| mean | Numeric mean value. Formatted to 1 decimal place.         |
| sd   | Numeric standard deviation. Formatted to 1 decimal place. |

**Value**

A character string of the form "X.X \u00b1 Y.Y" where both values are rendered to 1 decimal place using fixed-point notation.

---

|              |                                       |
|--------------|---------------------------------------|
| val_p_format | <i>Format a P value for reporting</i> |
|--------------|---------------------------------------|

---

**Description**

Format a P value for reporting

**Usage**

```
val_p_format(p, digits = 3)
```

**Arguments**

|        |                                                                                                                                                                                                       |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| p      | Numeric P value in the range [0, 1]. NA values are returned as NA_character_. Values >= 1 (or rounding to >= 1) are returned as e.g. ">0.999".                                                        |
| digits | Integer; number of decimal places for reported P values. Default is 3. Note: for p < 0.001, the value is reported in scientific notation with 1 significant figure regardless of digits (e.g., 8E-4). |

**Value**

A character string. Values < 0.001 are formatted in scientific notation with 1 significant figure (e.g., "8E-4"). All other values use fixed-point notation rounded to digits decimal places.

word\_export

*Export TernTables output to a formatted Word document***Description**

Export TernTables output to a formatted Word document

**Usage**

```
word_export(
  tbl,
  filename,
  round_intg = FALSE,
  round_decimal = NULL,
  font_size = 9,
  category_start = NULL,
  plain_header = NULL,
  subheader_rows = NULL,
  bold_rows = NULL,
  bold_sig = NULL,
  italic_rows = NULL,
  bold_cols = NULL,
  italic_cols = NULL,
  header_format_follow = FALSE,
  manual_italic_indent = NULL,
  manual_underline = NULL,
  table_caption = NULL,
  table_footnote = NULL,
  abbreviation_footnote = NULL,
  posthoc_footnote = NULL,
  variable_footnote = NULL,
  index_style = "symbols",
  page_break_after = FALSE,
  col1_header = NULL,
  line_break_header = getOption("TernTables.line_break_header", TRUE),
  open_doc = TRUE,
  citation = TRUE,
  font_family = getOption("TernTables.font_family", "Arial")
)
```

**Arguments**

|               |                                                                                                                                                   |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| tbl           | A tibble created by ternG or ternD                                                                                                                |
| filename      | Output file path ending in .docx                                                                                                                  |
| round_intg    | Logical; if TRUE, adds note about integer rounding. Default is FALSE.                                                                             |
| round_decimal | Integer or NULL; if provided, rounds all numeric values in the table to this many decimal places before rendering. Default is NULL (no rounding). |
| font_size     | Numeric; font size for table body. Default is 9.                                                                                                  |

|                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| category_start       | Named character vector specifying category headers. Names are header label text; values are anchor variable names – either the original column name or the cleaned display name (both forms accepted).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| plain_header         | Named character vector, same interface as category_start. Names are the label text to display; values are the anchor variable to insert before. The inserted row has text only in column 1 (all other cells blank) and receives <i>underline</i> formatting – identical to manual_underline – but no bold, merge, or border treatments. Default is NULL (none).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| subheader_rows       | Character vector of labels that already exist as rows in the table and should be formatted as full category section headers (merged across all columns, bold, with a bottom border line). Unlike category_start, no new row is inserted – the matching existing row is formatted in place. Intended for use with ternStyle() where section- divider rows are pre-built into the tibble. Case-insensitive match. Default NULL.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| bold_rows            | Integer vector of body row indices (1-based, in the final rendered table) to bold across every column. Applied as the last formatting pass so it overrides structural formatting. Default NULL.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| bold_sig             | <p>Optional named list for cell-level conditional bolding based on parsed p-values. Intended for use with ternStyle() when columns contain pre-formatted p-values that are not named "P value" (the name ternG uses internally). Supply a list with:</p> <ul style="list-style-type: none"> <li>• p_cols — character vector of column names containing p-value strings.</li> <li>• hr_cols — optional character vector of column names to also bold when the paired p-value is significant (must be the same length and order as p_cols, or NULL to skip paired HR bolding).</li> <li>• threshold — numeric significance threshold; default 0.05.</li> </ul> <p>For each p-value cell where the parsed numeric value is below threshold, that cell is bolded. If hr_cols is supplied, the corresponding HR cell in the same row is also bolded. The Variable column is never touched by this argument — use bold_rows to bold entire rows (e.g., predictor header rows). Default NULL.</p> |
| italic_rows          | Integer vector of body row indices to italicize across every column. Default NULL.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| bold_cols            | Integer vector of column indices (1-based) to bold across all body rows. Default NULL.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| italic_cols          | Integer vector of column indices to italicize across all body rows. Default NULL.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| header_format_follow | Logical; if TRUE, any columns listed in bold_cols or italic_cols also have their header cell bolded or italicized, respectively. Default FALSE.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| manual_italic_indent | Character vector of display variable names (post-cleaning) to force into italicized and indented formatting, matching the appearance of factor sub-category rows (e.g., levels of a multi-category variable). Use this for rows that should visually appear as sub-items but are not automatically detected as such.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| manual_underline     | Character vector of display variable names (post-cleaning) to force into underlined formatting, matching the appearance of multi-category variable header rows. Use this for rows that should visually appear as section headers but are not automatically detected as such.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| table_caption         | Optional character string to display as a caption above the table in the Word document. Rendered as size 11 Arial bold, single-spaced with a small gap before the table. Default is NULL (no caption).                                                                                                                                                                                                                                                                                                                                                                                                                           |
| table_footnote        | Optional character string to display as a footnote below the table in the Word document. Rendered as size 6 Arial italic. A double-bar border is applied above and below the footnote row. Default is NULL (no footnote).                                                                                                                                                                                                                                                                                                                                                                                                        |
| abbreviation_footnote | Optional character string (or character vector, which will be collapsed with spaces) listing abbreviations to display at the top of the footnote block. Always printed first, before any variable-specific footnote lines. Default NULL.                                                                                                                                                                                                                                                                                                                                                                                         |
| posthoc_footnote      | Optional character string describing post-hoc CLD superscript conventions. When supplied by <code>ternG()</code> , it is inserted after the abbreviations footnote and before the variable symbol footnotes. Default NULL.                                                                                                                                                                                                                                                                                                                                                                                                       |
| variable_footnote     | Optional named character vector. Names are display variable names as they appear in the table (case-insensitive match); values are the footnote definition text for that variable. Each entry is assigned the next symbol in the sequence (*, dagger, double-dagger, ...) and the symbol is appended to the variable name in column 1. The footnote block lists each as " <i>* Definition text.</i> " below the abbreviations. To map multiple variables to the same footnote symbol and note, separate the variable names with a pipe character in the key: <code>c("Var A Var B" = "Shared note text.")</code> . Default NULL. |
| index_style           | Character; controls the footnote symbol sequence. "symbols" (default) uses *, dagger, double-dagger, section, pilcrow, double-vertical-bar, then doubled forms. "*" is appended as plain text; all others are rendered as true Word superscripts. "alphabet" uses Unicode superscript letters (a, b, c, ...) which render as raised glyphs without explicit superscript formatting.                                                                                                                                                                                                                                              |
| page_break_after      | Logical; if TRUE, a page break is appended at the end of the Word document after the table. Used internally by <code>ternB()</code> to embed page breaks inside each table's temp file rather than injecting them into the combined document body, which avoids double-break artifacts when tables do not fill the page. Default is FALSE.                                                                                                                                                                                                                                                                                       |
| col1_header           | Optional character string. Overrides the top-left header cell text. When NULL (default), the cell shows "Category\n Variable" (the standard two-line label). Supply any string, including "\n" line breaks, to customise. Example: "Variable\n Index Management Strategy".                                                                                                                                                                                                                                                                                                                                                       |
| line_break_header     | Logical; if TRUE (default), column headers are wrapped with \n – group names break on spaces, sample size counts move to a second line, and the first column header includes a category hierarchy label. Set to FALSE to suppress all header line breaks. Can also be set package-wide via <code>options(TernTables.line_break_header = FALSE)</code> .                                                                                                                                                                                                                                                                          |
| open_doc              | Logical; if TRUE (default), automatically opens the written Word document in the system default application after saving. Set to FALSE to suppress.                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| citation              | Logical; if TRUE (default), appends a citation line at the bottom of the table footnote block: package version, authors, and links to the GitHub repository and web interface. Set to FALSE to suppress.                                                                                                                                                                                                                                                                                                                                                                                                                         |

**font\_family** Character; font family used for the entire Word table and its caption, footnote, and citation. Any font name accepted by the rendering system is valid (Word will fall back to its default if the font is not installed). Can also be set package-wide via `options(TernTables.font_family = "Garamond")`. Default is "Arial".

### Value

Invisibly returns the path to the written Word file.

### Examples

```
data(tern_colon)
tbl <- ternD(tern_colon, exclude_vars = c("ID"), methods_doc = FALSE, open_doc = FALSE)
word_export(
  tbl      = tbl,
  filename = file.path(tempdir(), "descriptive.docx"),
  open_doc = FALSE,
  category_start = c(
    "Patient Demographics" = "Age (yr)",
    "Tumor Characteristics" = "Positive Lymph Nodes (n)"
  )
)
```

---

|                    |                                                           |
|--------------------|-----------------------------------------------------------|
| write_cleaning_doc | <i>Write a cleaning summary document for ternP output</i> |
|--------------------|-----------------------------------------------------------|

---

### Description

Generates a Word document summarising the preprocessing transformations applied by [ternP](#). Only sections for triggered transformations are written; if the data required no preprocessing, a single sentence stating that is produced instead. The document can be attached to a data-management log or supplemental materials.

### Usage

```
write_cleaning_doc(
  result,
  filename = "cleaning_summary.docx",
  font_family = getOption("TernTables.font_family", "Arial"),
  open_doc = TRUE,
  citation = TRUE
)
```

### Arguments

|                    |                                                                                                                                                       |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>result</b>      | A <code>ternP_result</code> object returned by <a href="#">ternP</a> .                                                                                |
| <b>filename</b>    | Output file path ending in <code>.docx</code> . Default is <code>"cleaning_summary.docx"</code> in the current working directory.                     |
| <b>font_family</b> | Character; font family for the Word document. Default <code>"Arial"</code> . Can also be set via <code>options(TernTables.font_family = ...)</code> . |

|          |                                                                                                                                                     |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| open_doc | Logical; if TRUE (default), automatically opens the written Word document in the system default application after saving. Set to FALSE to suppress. |
| citation | Logical; if TRUE (default), appends the TernTables citation as a page footer in the written document. Set to FALSE to suppress.                     |

### Value

Invisibly returns the path to the written Word file.

### See Also

[ternP](#), [write\\_methods\\_doc](#)

### Examples

```
path <- system.file("extdata/csv", "tern_colon_messy.csv",
                    package = "TernTables")
raw <- read.csv(path, stringsAsFactors = FALSE)
result <- ternP(raw)
write_cleaning_doc(result, filename = file.path(tempdir(), "cleaning_summary.docx"),
                  open_doc = FALSE)
```

---

write\_methods\_doc

*Write a methods section Word document for TernTables output*

---

### Description

Generates a Word document containing a methods paragraph describing the statistical approach used in a specific `ternG` or `ternD` run. The paragraph is fully dynamic: it reflects the tests that were actually used, the number of comparison groups, whether odds ratios were calculated, and whether post-hoc testing was performed. It is headed by a bold **Statistical Methods** label and followed by a brief attribution footer.

### Usage

```
write_methods_doc(
  tbl,
  filename,
  n_levels = 2,
  OR_col = FALSE,
  OR_method = "dynamic",
  source = "ternG",
  post_hoc = FALSE,
  categorical_posthoc = FALSE,
  cat_posthoc_fisher_vars = character(0),
  show_missingness = FALSE,
  missing_indicators = NULL,
  boilerplate = FALSE,
  p_adjust = FALSE,
  open_doc = TRUE,
  citation = TRUE,
```

```
font_family = getOption("TernTables.font_family", "Arial")
)
```

## Arguments

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| tbl                     | A tibble created by <code>ternG</code> or <code>ternD</code> , or <code>NULL</code> when generating a generic document.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| filename                | Output file path ending in <code>.docx</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| n_levels                | Number of group levels used in <code>ternG</code> (2 for two-group, 3+ for multi-group). Ignored when called from <code>ternD</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| OR_col                  | Logical; whether odds ratios were calculated. Default <code>FALSE</code> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| OR_method               | Character; the OR calculation method used in <code>ternG</code> . "dynamic" (default) means Wald when all expected cells $\geq 5$ , Fisher's exact otherwise. "wald" means Wald was forced regardless of cell counts. Controls the OR description in the generated methods paragraph.                                                                                                                                                                                                                                                                                                                      |
| source                  | Character; "ternG" or "ternD". Controls which section is populated with dynamic test information. Default "ternG".                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| post_hoc                | Logical; whether pairwise post-hoc testing was requested ( <code>post_hoc = TRUE</code> in <code>ternG</code> ). When <code>TRUE</code> and <code>n_levels</code> $\geq 3$ , the three-group methods paragraph is updated to describe the post-hoc test pairing (Games-Howell or Dunn's + Holm). Default <code>FALSE</code> .                                                                                                                                                                                                                                                                              |
| categorical_posthoc     | Logical; whether adjusted standardized residuals were requested ( <code>categorical_posthoc = TRUE</code> in <code>ternG</code> ). When <code>TRUE</code> and <code>n_levels</code> $\geq 3$ , the methods paragraph notes that cells with adjusted standardized residuals exceeding $\pm 1.96$ are marked with an asterisk following a significant omnibus test. Default <code>FALSE</code> .                                                                                                                                                                                                             |
| cat_posthoc_fisher_vars | Character vector of variable names for which Fisher's exact test was the omnibus test while <code>categorical_posthoc = TRUE</code> . When non-empty, a caveat sentence is appended noting that Haberman's adjusted residuals were derived from the chi-squared contingency table in the absence of a Fisher's exact equivalent. Populated automatically when called from <code>ternG()</code> . Default <code>character(0)</code> .                                                                                                                                                                       |
| show_missingness        | Logical or character; whether missingness columns were added to the table ( <code>FALSE</code> , "total", or "group"). When non- <code>FALSE</code> , a sentence is appended describing the missingness reporting approach and the string representations used to flag missing values. Should match the <code>show_missingness</code> argument passed to <code>ternG()</code> or <code>ternD()</code> . Default <code>FALSE</code> .                                                                                                                                                                       |
| missing_indicators      | Character vector of string values treated as missing in addition to R <code>NA</code> , or <code>NULL</code> to use <code>TernTables</code> defaults. Should match the <code>missing_indicators</code> argument passed to <code>ternG()</code> or <code>ternD()</code> . Default <code>NULL</code> .                                                                                                                                                                                                                                                                                                       |
| boilerplate             | Logical; if <code>TRUE</code> , ignores all other arguments and writes a single comprehensive Word document covering every possible <code>TernTables</code> configuration (descriptive, two-group with and without odds ratios, three-or-more-group with and without post-hoc testing), using package-default phrasing throughout. Output is written to <code>filename</code> if supplied; otherwise falls back to <code>comprehensive_boilerplate_methods.docx</code> in the current working directory. Intended as a reference document, not for inclusion in a manuscript. Default <code>FALSE</code> . |



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